NON-WSQ COURSE

LESSON PLAN

For

Arduino Masterclass

Course Code: C218

Conducted by Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

Document Version Control Record

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1.0	30 September 2026	First version	Tertiary Infotech Academy Pte Ltd

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Course Information

Course title	Arduino Masterclass
Course code	C218
Duration	7.5 instructional hours (one day, 9:30am - 5:30pm)
Mode	Instructor-led, demonstration, didactic questioning, and hands-on practice

Learning Outcomes

- LO1 - Analyse real-life applications using Arduino microcontroller
- LO2 - Implement basic electrical circuits using Arduino microcontroller
- LO3 - Apply Arduino microcontroller to analogue sensors and transducers
- LO4 - Apply Arduino microcontroller to digital sensors and transducers
- LO5 - Implement actuators with Arduino microcontroller

One-Day Schedule

Time	Minutes	Type	Facilitator / learner activity	Resources / slide reference
09:30-09:55	25	**Admin**	Trainer/learner introduction, outcomes and safety	Slides 1-9
09:55-10:40	45	**Topic**	Topic 1: Introduction to Embedded Systems	Slides 10-27
10:40-11:15	35	**Lab**	Labs 1-2: board/serial evidence	Lab folders 01-02
11:15-12:05	50	**Topic**	Topic 2: Introduction to Basic Electronics	Slides 28-46
12:05-12:50	45	**Lab**	Labs 3-5: safe LED, button, PWM	Lab folders 03-05
12:50-13:20	30	**Break**	Lunch break	-
13:20-14:05	45	**Topic**	Topic 3: Analogue Sensors and Transducers	Slides 47-64
14:05-14:40	35	**Lab**	Labs 6-7: LDR and servo	Lab folders 06-07
14:40-15:25	45	**Topic**	Topic 4: Digital Sensors and Transducers	Slides 65-83
15:25-15:55	30	**Lab**	Labs 8-9: ultrasonic and PIR	Lab folders 08-09

Time	Minutes	Type	Facilitator / learner activity	Resources / slide reference
15:55-16:40	45	**Topic**	Topic 5: Actuators	Slides 84-102
16:40-17:15	35	**Lab**	Labs 10-12: DHT, stepper and fan control	Lab folders 10-12
17:15-17:30	15	**Review**	Verification matrix, course recap and feedback	Slides 103-106

Total instructional time: 450 minutes (7.5 hours). Lunch: 30 minutes. Full day: 480 minutes, 09:30-17:30.

Alignment Matrix

Outcome	Skills and knowledge	Slides	Labs	Evidence
LO1: Analyse real-life applications using Arduino microcontroller	Execute performance specification and analysis of sensors and actuators for real-life applications Underlying concepts pertaining to performance specification and analysis	Slides 10-27	Labs 1, 2	Lab acceptance tests
LO2: Implement basic electrical circuits using Arduino microcontroller	Utilise commonly adopted component interconnections and signal-conditioning principles in automation Implementation of component interconnections and signal conditioning concepts	Slides 28-46	Labs 3, 4, 5	Lab acceptance tests
LO3: Apply Arduino microcontroller to analogue sensors and transducers	Apply analogue sensors and transducers to solve real-world control problems Applications of analogue sensors and transducers	Slides 47-64	Labs 6, 7	Lab acceptance tests
LO4: Apply Arduino microcontroller to digital sensors and transducers	Apply digital transducers to solve real-world control problems Applications of digital transducers	Slides 65-83	Labs 8, 9	Lab acceptance tests
LO5: Implement actuators with Arduino microcontroller	Develop actuator networks with stepper and continuous-drive actuators Actuator networks	Slides 84-102	Labs 10, 11, 12	Lab acceptance tests

Facilitator Guidance

- Keep click-by-click procedures in this Learner Guide and lab READMEs. Use slides to explain mechanism, evidence, decision rules, and failure controls.
- Before powering a circuit, ask learners to trace the complete current path and identify the energy boundary.
- Before running code, ask learners to predict input, state transition, output, and acceptance evidence.
- For every lab, collect at least one observable artifact: serial trace, timing measurement, ADC range, state result, schematic, or prototype evidence.